



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS**

**A PROPOSAL
ON
MOTIONBRIDGE: A MULTI-CAMERA NEURAL NETWORK FRAMEWORK FOR
REAL-TIME HUMAN ARM MOTION REPLICATION AND DATASET GENERATION**

Submitted By:

Pratik Khatiwada (THA079BEI025)
Preeti Rijal (THA079BEI028)
Sampada Ghimire (THA079BEI036)

Submitted To:

Department of Electronics and Computer Engineering
Thapathali Campus
Kathmandu, Nepal

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Pratik Khatiwada (THA079BEI025)

Preeti Rijal (THA079BEI028)

Sampada Ghimire (THA079BEI036)

ABSTRACT

MotionBridge is a low-cost, real-time, multi-camera framework designed for human arm motion capture, simulation replication, and robotic dataset generation without relying on expensive depth sensors or wearable hardware. Utilizing three standard, calibrated RGB webcams, the system extracts 2D joint coordinates and 21-point hand landmarks per frame using MediaPipe Pose and MediaPipe Hands. These 2D keypoints are fused via a confidence-weighted triangulation algorithm into accurate 3D metric world coordinates. The reconstructed positions are mapped in real time to a simulated robotic arm inside a PyBullet physics environment using Inverse Kinematics (IK). Operating efficiently on standard CPU hardware, the pipeline simultaneously renders a live visual replication and exports a structured CSV dataset containing 3D joint trajectories, computed angles, and timestamps to support downstream robotic imitation learning.

Keywords: *Motion Capture, Multi-Camera Triangulation, MediaPipe, PyBullet Simulation, Inverse Kinematics, Robotic Dataset Generation, Computer Vision, Real-Time Systems.*

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LIST OF ABBREVIATIONS

2D	Two-Dimensional
3D	Three-Dimensional
CPU	Central Processing Unit
CSV	Comma-Separated Values
DRL	Deep Reinforcement Learning
FPS	Frames Per Second
GPU	Graphics Processing Unit
HPE	Human Pose Estimation
HRC	Human-Robot Collaboration
IK	Inverse Kinematics
IMU	Inertial Measurement Unit
NPR	Nepalese Rupee
RGB	Red Green Blue
RGB-D	Red Green Blue - Depth
SVD	Singular Value Decomposition
URDF	Unified Robot Description Format
USB	Universal Serial Bus
VAR	Video Assistant Referee
VLM	Vision-Language Model

1. INTRODUCTION

1.1. Background

The ability to capture, interpret, and replicate human motion has long been a fundamental challenge in robotics, computer vision, and human-computer interaction. Traditional motion capture systems rely on expensive marker-based setups, wearable Inertial Measurement Unit (IMU), or specialized depth sensors such as the Microsoft Kinect and Intel RealSense, all of which impose significant hardware constraints and deployment costs. Recent advances in deep learning-based pose estimation, particularly lightweight frameworks such as MediaPipe developed by Google, have enabled accurate real-time extraction of human joint coordinates from standard RGB video without specialized hardware.

Simultaneously, physics-based simulation platforms such as PyBullet have matured into robust environments for replicating articulated body motion, providing physically accurate rendering of joint movement and a foundation for generating structured motion datasets. The convergence of these two technologies vision-based pose estimation and physics-based simulation presents an opportunity to build affordable, accessible motion capture systems that require only standard cameras.

In the context of industrial robotics and assistive automation, large-scale datasets of human arm movement are essential for training robotic manipulators to perform tasks mimicking natural human motion. However, the cost and complexity of existing data collection pipelines remain a barrier to widespread dataset generation. A vision-based, multi-camera system capable of capturing full arm-to-fingertip motion in real-time and replicating it in simulation addresses this barrier directly.

1.2. Motivation

Robotic arms capable of replicating human arm motion have significant potential in industrial automation, assistive robotics, teleoperation, and rehabilitation engineering. A core requirement for training such systems is access to large, diverse datasets of human arm trajectories annotated with joint coordinates and angles. Existing approaches to generating such datasets either rely on expensive hardware (RGB-D sensors, IMU suits) or produce offline, non-real-time outputs that are difficult to scale.

The motivation for MotionBridge arises from the absence of a low-cost, real-time, vision-only pipeline that combines multi-camera capture, neural network-based pose estimation, three-dimensional reconstruction, and physics simulation into a single integrated system. By using only standard USB webcams and open-source software, the proposed system can be deployed in resource-constrained environments such as

academic laboratories, small research groups, and developing-world institutions where expensive motion capture infrastructure is unavailable.

Furthermore, no existing reviewed work captures both full arm joint trajectories and detailed finger joint coordinates (21 points per hand) from standard RGB cameras and simultaneously replicates arm movement in simulation. MotionBridge addresses this gap, producing a richer dataset than currently available vision-based approaches.

1.3. Problem Statement

Existing human arm motion capture systems for robotic dataset generation suffer from one or more of the following limitations: reliance on expensive depth-sensing hardware, dependence on wearable sensors, absence of real-time simulation replication, operation on monocular camera inputs that cannot resolve depth accurately, or omission of finger-level joint data. These constraints prevent scalable and affordable motion dataset generation for robotic arm training.

This project addresses the problem of building a low-cost, real-time system that captures three-dimensional human arm and finger joint motion using only standard RGB cameras, replicates the captured motion in a physics-based simulation environment, and produces a structured dataset of joint coordinates and angles for downstream robotic arm training without requiring specialized depth sensors, wearable hardware, or GPU-accelerated inference.

1.4. Objectives

- To implement a vision-only system that extracts real-time 2D arm and hand landmarks from three standard RGB webcams and fuses them into 3D world coordinates via confidence-weighted triangulation.
- To map reconstructed 3D joint trajectories onto a simulated PyBullet robotic arm using Inverse Kinematics while simultaneously exporting a structured CSV dataset for robotic training.

1.5. Project Scope

1.5.1. Capabilities

- Real-time capture of human arm movement (shoulder, elbow, wrist) using three standard USB webcams.
- Detection of 21 finger joint landmarks per hand using MediaPipe Hands in addition to full arm keypoints.

- Three-dimensional joint coordinate reconstruction via confidence-weighted triangulation using calibrated multi-camera geometry.
- Real-time replication of arm movement in PyBullet physics simulation using Inverse Kinematics.
- Generation of a structured dataset containing per-frame 3D joint coordinates, joint angles, and timestamps.
- Occlusion robustness through multi-camera redundancy when one camera's view is blocked, remaining cameras compensate.

1.5.2. Limitations

- Finger joint movement is captured and recorded in the dataset but is not replicated in PyBullet simulation, as standard arm URDF models do not include articulated finger structures.
- The system is limited to single-person capture; multi-person simultaneous tracking is outside scope.
- Accuracy depends on initial camera calibration quality; dynamic camera repositioning requires recalibration.
- The system does not include robot arm training or policy learning; it generates the dataset only.
- Performance may degrade under extreme lighting conditions or significant self-occlusion not resolvable by three cameras.
- The system does not support full-body motion capture only the arm from shoulder to fingertip.

1.6. Problem Applications

The dataset and simulation output produced by MotionBridge have direct applications in the following domains:

- **Industrial Automation:** Providing training data for robotic arms performing labour-intensive factory tasks such as assembly, sorting, and load-carrying that mimic human arm motion.
- **Assistive Robotics:** Generating motion trajectories for robotic prosthetics and assistive devices that replicate natural human arm and hand movement.
- **Teleoperation:** Serving as a reference system for mapping human operator arm motion to remote robotic manipulators.
- **Rehabilitation Engineering:** Capturing and analysing patient arm movement for physiotherapy assessment and progress monitoring.

- **Imitation Learning Research:** Supplying labelled demonstration datasets for behavioral cloning and inverse reinforcement learning pipelines.
- **Simulation-Based Training:** Using the PyBullet output as a reference motion environment for reinforcement learning agents.

2. LITERATURE REVIEW

2.1. Related Works

Human Pose Estimation (HPE) forms the foundational stage of any vision-based motion capture system. Kim et al. [1] demonstrated the effectiveness of MediaPipe Pose in a two-stage pipeline running on a single-board computer mounted on a mobile robot, achieving a mean joint coordinate difference of 0.097 m and an average angle difference of 10.017 degrees per joint across six daily poses. The study confirmed that MediaPipe provides a practical balance between computational cost and accuracy for real-time embedded applications, directly supporting its adoption as the pose estimation backbone in the proposed system.

Sarkar et al. [2] implemented a multi-camera system for close-proximity human-robot collaboration in construction environments, using MediaPipe Pose to extract 15 keypoints per frame from each camera view and converting them to 3D coordinates through rotation, translation, and homogeneous transformation matrices derived from known camera positions and orientations. This work is the closest architectural match to the proposed pipeline, validating the feasibility of combining MediaPipe with a multi-camera setup for real-world 3D joint reconstruction without specialized depth hardware.

Zell et al. [3] presented a multimodal human motion dataset containing 51,051 poses from 71 persons, obtained by extracting 2D keypoints using MediaPipe Human Pose Landmarker from virtual multi-camera images and computing corresponding 3D keypoints by linear triangulation. Their dataset was explicitly intended for transforming computer vision pose solutions into biomechanical simulations, constituting a direct proof of concept for the MediaPipe-plus-triangulation pipeline proposed in MotionBridge.

Bultmann and Behnke [4] proposed a distributed multi-camera architecture where 2D joint detection is performed locally at each smart edge sensor and only the semantic skeleton representation is transmitted to a central backend for 3D triangulation. Their system achieves real-time 3D pose estimation while significantly reducing network bandwidth, and incorporates prior human skeleton knowledge into the triangulation backend to improve robustness an architectural approach directly relevant to the distributed and real-time requirements of the proposed system.

Bragagnolo et al. [5] addressed the known limitation of standard triangulation under occlusion by performing multi-view fusion on 3D skeletons from monocular methods rather than fusing noisy 2D features, applying reprojection error optimization with limb-length symmetry constraints. Their method significantly outperformed standard

triangulation in human-robot collaboration scenarios with heavy occlusion, identifying a clear accuracy upgrade path for the proposed system beyond its baseline triangulation implementation.

Leroy et al. [6] extended multi-view 3D pose estimation to uncalibrated camera setups through Probabilistic Triangulation, using Monte Carlo sampling to iteratively estimate camera pose distributions. Their experiments on Human3.6M and CMU Panoptic benchmarks demonstrated that accurate 3D reconstruction is achievable without fixed camera calibration, which is relevant when the proposed system is deployed in environments where precise camera recalibration is impractical.

Liang et al. [7] proposed the most directly related simulation-based work, integrating 3D HPE with PyBullet for robotic arm motion imitation. Their pipeline extracts human arm motion from video using a Strided Transformer network, retargets the human morphology to a robot URDF model, generates reference trajectories in PyBullet using its built-in Inverse Kinematics solver, and applies deep reinforcement learning for policy training. However, their system operates offline on monocular pre-recorded video and targets robot policy training rather than dataset generation, distinguishing it from the real-time multi-camera focus of MotionBridge.

Jiang et al. [8] demonstrated real-time multi-person arm motion imitation using a single RGB-D camera and an improved retargeting algorithm, achieving consistent arm configuration and precise end-effector tracking for multi-robot collaboration tasks. While their system confirms that real-time arm motion replication is feasible, it relies on expensive RGB-D hardware for depth acquisition, whereas MotionBridge targets equivalent real-time performance using only standard RGB cameras with computational triangulation, substantially lowering the deployment cost.

Faria et al. [9] presented a teleoperation framework combining MediaPipe landmark detection with an Intel RealSense RGB-D camera for 3D joint reconstruction, mapping the results to a robot arm’s kinematic constraints in real-time. Their work explicitly noted that vision-only estimation remains limited for axial rotations such as elbow and wrist yaw under occlusions a limitation that the multi-camera triangulation approach in MotionBridge is specifically designed to mitigate by providing redundant viewpoints.

Moya-Alcover et al. [10] presented seven motion capture datasets of professional industrial gestures performed by skilled craftsmen and factory operators, recorded using wearable inertial sensors. Their work is motivated by the same industrial and labour application domain described in MotionBridge capturing arm motion for tasks involving factory work and load-carrying but relies on instrumented suits rather than

vision-based capture, introducing hardware constraints that the proposed camera-based system avoids entirely.

2.2. Existing Systems and Limitations

A comparative analysis of the reviewed works reveals consistent gaps across three dimensions: hardware cost, depth resolution method, and system completeness. Table 2-1 summarises the key attributes of existing systems against the proposed MotionBridge system.

Table 2-1. Comparison of Related Works with Proposed System

Work	Year	Camera Setup	Depth Method	Simulation	Primary Goal
Liang et al.	2024	Monocular	Transformer	PyBullet	DRL Training
Jiang et al.	2025	Single RGB-D	Hardware	Real Robot	Robot Control
Faria et al.	2025	Single RGB-D	Hardware	Real Robot	Teleoperation
Sarkar et al.	2022	Multi RGB	Homogeneous	None	HRC Safety
Bragagnolo et al.	2024	Multi RGB	Multi-view Fusion	None	Pose Accuracy
Bultmann & Behnke	2021	Multi (edge)	Triangulation	None	Real-time 3D
Moya-Alcover et al.	2023	IMU-based	Inertial	None	Dataset
Zell et al.	2024	Virtual Multi	Triangulation	None	Dataset
MotionBridge	2026	Multi RGB	Conf.-Weighted Tri.	PyBullet	Simulation + Dataset

Systems that achieve real-time arm replication (Jiang et al., Faria et al.) depend on hardware depth sensors costing \$300 or more. Systems using standard cameras and triangulation (Sarkar et al., Zell et al., Bultmann and Behnke) omit the simulation replication stage entirely. The only system using PyBullet for arm motion replication (Liang et al.) operates offline on monocular input and produces no structured dataset. No reviewed work captures finger-level joint data alongside arm joints from standard RGB cameras. MotionBridge uniquely combines all four capabilities multi-camera RGB capture, real-time pose estimation, confidence-weighted triangulation, PyBullet simulation replication, and full arm-to-fingertip dataset generation in a single low-cost pipeline.

3. REQUIREMENTS

3.1. Hardware Requirements

- **USB Webcams × 3:** Standard 1080p USB webcams (e.g., Logitech C270 or equivalent), positioned at front, left, and right angles around the capture area. Minimum 30 fps required for real-time operation.
- **Camera Mounts / Tripods × 3:** Fixed mounting hardware to ensure stable, calibrated camera positions during capture sessions.
- **Checkerboard Calibration Target:** Printed A4 checkerboard pattern (9×6 inner corners) for OpenCV camera calibration.
- **Laptop / Desktop PC:** Minimum Intel Core i5 (8th generation or above), 8 GB RAM. No GPU required. MediaPipe and PyBullet both run on CPU for this pipeline.
- **USB Hub (optional):** If the host machine has insufficient USB ports for three simultaneous camera connections.

3.2. Software Requirements

- **Programming Language:** Python 3.9 or above.
- **Pose Estimation:** MediaPipe 0.10+ (`mediapipe` Python package) - for real-time arm and hand landmark detection.
- **Computer Vision:** OpenCV 4.8+ (`opencv-python`) - for camera capture, calibration, and triangulation.
- **Physics Simulation:** PyBullet (`pybullet`) - for IK computation and real-time arm simulation rendering.
- **Numerical Computing:** NumPy (`numpy`) - for matrix operations in triangulation and IK.
- **Data Storage:** Python `csv` module or Pandas (`pandas`) - for structured dataset writing.
- **Development Environment:** Visual Studio Code on Windows 10/11 or Ubuntu 22.04.
- **Version Control:** Git / GitHub - for source code management.

4. PROPOSED SYSTEM/ METHODOLOGY

4.1. System Overview

The MotionBridge framework is designed as a modular, hardware-agnostic software pipeline that translates real-time multi-view human upper limb movements directly into physical simulation joint trajectories, automating behavioral cloning dataset factory operations. The structural design isolates computer vision feature extraction from physical joint simulation and downstream time-series machine learning policies. The high-level static distribution of components, interfaces, and boundary layers across the three primary execution phases of the framework is illustrated in Figure 4-1.

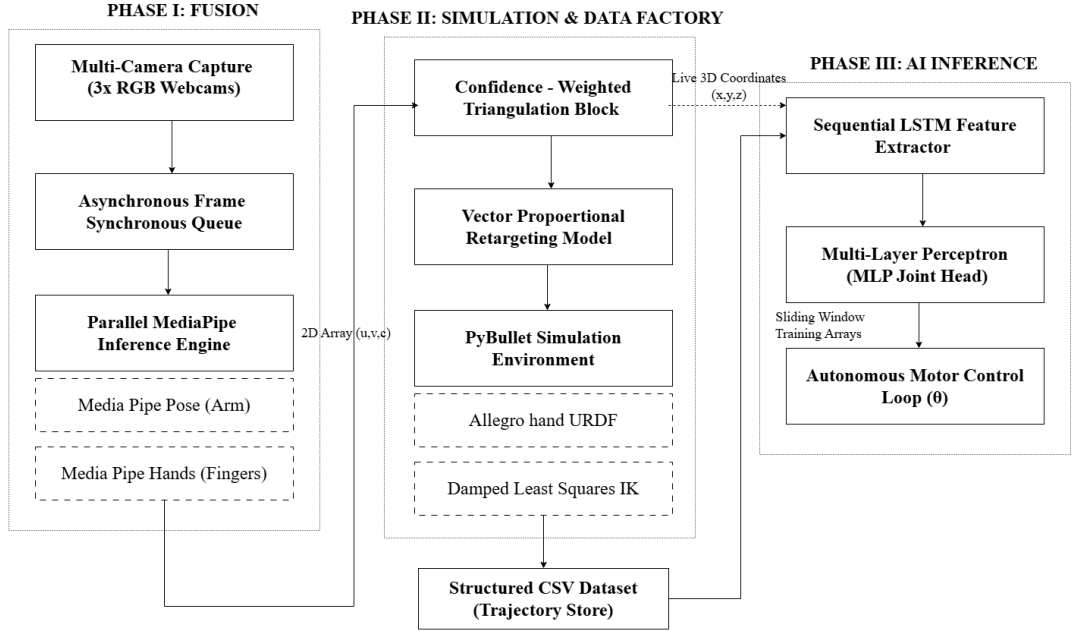


Figure 4-1. Block diagram of the MotionBridge framework.

4.1.1. Phase I: Multi-View Spatial Data Acquisition & Fusion

Geometric Camera Calibration

To eliminate lens distortion artifacts and define structural world space mapping variables, three consumer-grade RGB 1080p webcams are fixed on rigid tripods encircling the operator workspace. Geometric evaluation relies on OpenCV's checkerboard target matrix configuration (9×6 internal corners). The algorithm computes the individual camera intrinsic parameter matrix K_i and the camera extrinsic spatial orientation matrix $[R_i|t_i]$, defining a permanent structural projection matrix P_i per camera view i :

$$P_i = K_i \cdot [R_i|t_i] \quad \text{where } i \in \{1, 2, 3\} \quad (4-1)$$

Multi-Threaded Frame Synchronization

Asynchronous USB bus clock jitters create temporal variations across video streams, compromising spatial triangulation. To mitigate this issue, individual camera captures are isolated onto separate thread loops managed via a central thread queue. Incoming frames are assigned system arrival timestamps (t_{cam}); a matching mechanism drops unaligned frame pairs, enforcing temporal consistency across views if and only if:

$$\Delta t = |t_{\text{cam},i} - t_{\text{cam},j}| < 15 \text{ ms} \quad \forall i, j \in \{1, 2, 3\} \quad (4-2)$$

Parallel Multi-Model Keypoint Extraction

Synchronized frames are concurrently transmitted to localized, independent instances of Google MediaPipe libraries. The frame processing loop targets parallel data extraction tasks:

1. **MediaPipe Pose Landmarker:** Extracts structural coordinates defining large-scale body vectors, focusing on the shoulder, elbow, and wrist nodes.
2. **MediaPipe Hands Mesh Model:** Tracks dense regional features by mapping 21 localized tracking landmarks per hand to monitor intricate gesturing profiles.

This combined parsing generates a continuous stream of two-dimensional pixel arrays $x_i = (u_i, v_i)$ paired with an associated, dynamic visibility validation metric $c_i \in [0, 1]$.

Confidence-Weighted Triangulation Calculus

Standard linear triangulation using Singular Value Decomposition ($AX = 0$) breaks down under severe occlusion conditions common to single-user movement. MotionBridge counteracts viewpoint dropouts by adjusting the projection matrix influence dynamically using MediaPipe's live visibility score (c_i). This calculation weights down occluded views, computing accurate 3D metric world coordinates (X_{3D}):

$$X_{3D} = \frac{\sum_{i=1}^3 c_i \cdot f(P_i, x_i)}{\sum_{i=1}^3 c_i} \quad (4-3)$$

4.1.2. Phase II: Simulation Mapping, Kinematic Processing, & Automated Dataset Generation

Vector Proportional Retargeting

Because human bone proportions vary significantly from robot dimensions, raw metric world values (X_{3D}) cannot be directly mapped to joint targets. To address this, human tracking segments are evaluated as directional spatial vectors:

$$\vec{V}_{\text{segment}} = X_{3D, \text{child}} - X_{3D, \text{parent}} \quad (4-4)$$

These vectors are proportionally normalized and scaled according to the mechanical link lengths specified within the robot URDF asset file, preventing joint hyper-extension and structural calculation drift.

Articulated Kinematic Setup within PyBullet

The simulation environment is built inside the PyBullet physics engine utilizing the Kuka IIWA URDF manipulator model. This environment provides physically accurate rendering of joint movements and maps the mathematical spatial features to a standardized robotic configuration.

Damped Least Squares Inverse Kinematics

To map target positions to joint rotations without encountering mathematical divisions by zero near arm workspace boundaries (singularities), PyBullet's internal solver is configured to evaluate joint states via a Damped Least Squares (DLS) optimization algorithm:

$$\theta = \text{pybullet.calculateInverseKinematics}(\text{bodyId}, \text{jointId}, X_{\text{wrist}}) \quad (4-5)$$

The calculated joint configuration array θ updates the virtual robot model at a consistent execution rate via motor position control commands.

Automated Data Factory Serialization

While maintaining live display rendering, the system operates as an automated data logging factory. The processing loop writes synchronized parameters directly to a structured CSV format trajectory file. Each line records the spatial features and target joint angles for that frame:

$$\text{Record} = \{t, X_{\text{shoulder}}, X_{\text{elbow}}, X_{\text{wrist}}, X_{\text{fingers}}, \theta\} \quad (4-6)$$

4.1.3. Phase III: Downstream Behavioral Cloning Policy (The AI Network)

Temporal Sliding Window Reshaping

To train the machine learning framework to capture structural motion characteristics rather than static positions, the stored CSV logs are prepared for training using a sliding window strategy. Rows are reshaped into sequential arrays spanning 10 consecutive frames:

$$\mathcal{S}_t = [\text{Record}_{t-9}, \text{Record}_{t-8}, \dots, \text{Record}_t] \quad (4-7)$$

This temporal structure allows the network to learn higher-order motion parameters, such as velocity and trajectory acceleration vectors.

LSTM-MLP Architecture and Policy Optimization

The prediction model is built in PyTorch using a sequential architecture:

- **LSTM Layer Block:** Processes time-series state sequences ($\mathcal{S}_t$) to model historical trajectory paths and movement dynamics.
- **MLP Target Head:** Takes the final hidden state vector from the LSTM and passes it through fully connected layers to predict the complete array of target motor angles ($\theta_{\text{predicted}}$).

The policy weights are optimized by minimizing Mean Squared Error (MSE) loss relative to the analytical DLS outputs collected during Phase II:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{j=1}^N (\theta_{\text{predicted},j} - \theta_{\text{analytical},j})^2 \quad (4-8)$$

4.1.4. Deployment Framework: The Autonomous Control Loop

Once network optimization converges, the analytical Damped Least Squares Inverse Kinematics solver is decoupled and disabled. The live runtime script switches to an autonomous, forward-pass inference model. Raw webcam data passes through localized 2D extraction and triangulation math, routing directly into the trained LSTM-MLP policy engine to output joint angles (θ) instantly. This eliminates the need for iterative matrix calculus, reducing system overhead and sustaining real-time processing speeds of 25–30 FPS on consumer CPUs. The step-by-step runtime pipeline execution path is illustrated in Figure 4-2.

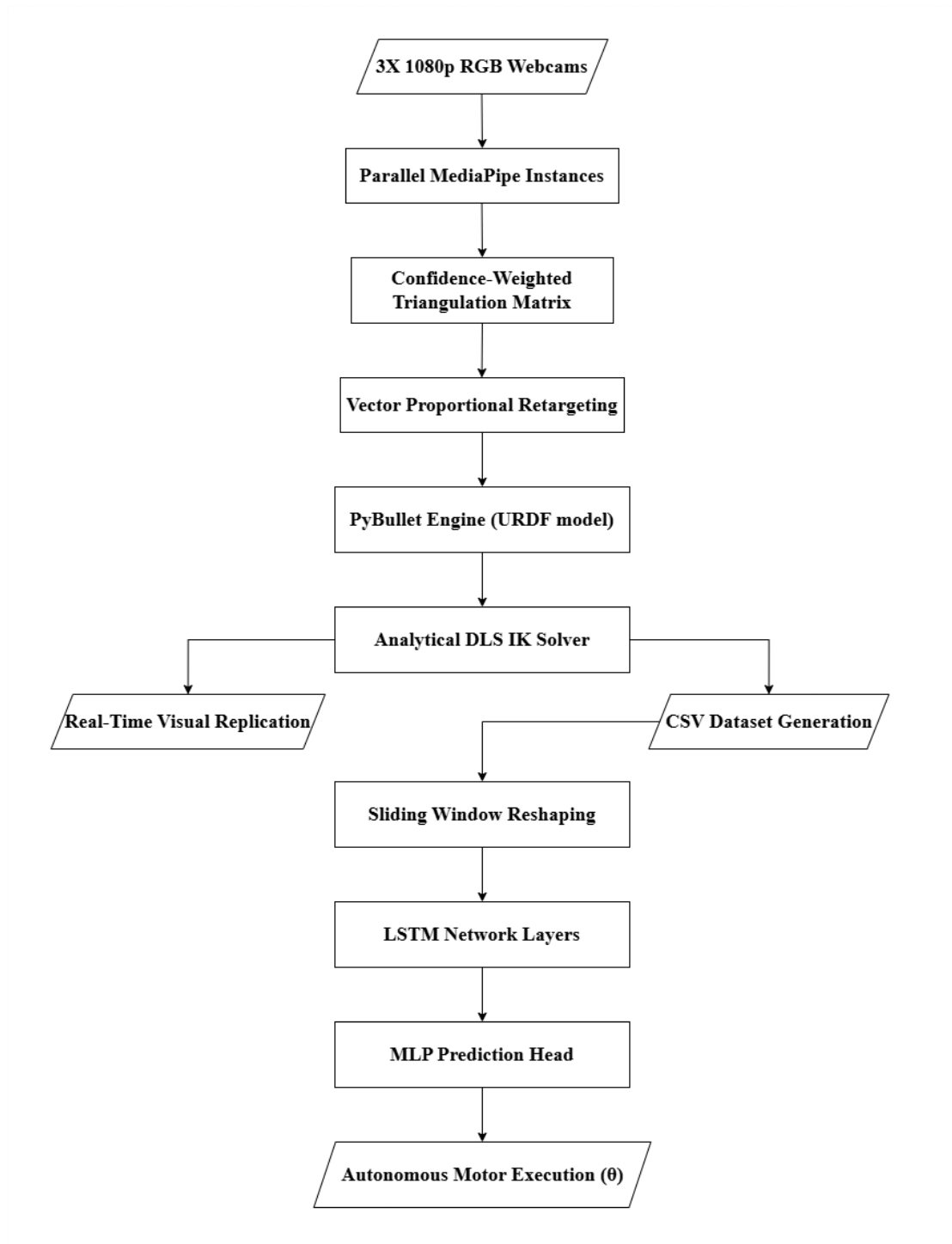


Figure 4-2. Procedural Flowchart outlining the runtime sequence of software logic execution from initial frame acquisition to dataset training and control output.

5. EXPECTED OUTCOMES

The primary deliverable of MotionBridge is a functioning real-time software system that accepts input from three synchronized USB webcams and produces two simultaneous outputs: a live PyBullet simulation window showing the simulated arm replicating the human subject’s arm movement, and a continuously written CSV dataset file containing per-frame 3D joint coordinates and joint angles.

The expected dataset will contain full arm joint data (shoulder, elbow, wrist) and 21-point finger joint data per hand at the capture frame rate (target: 25–30 fps), with 3D joint position accuracy within acceptable bounds for downstream imitation learning pipelines. The system is expected to maintain real-time performance on standard laptop hardware without GPU acceleration.

Key performance metrics to be evaluated include: mean 3D joint reconstruction error (compared against a ground-truth reference pose), frames-per-second throughput of the full pipeline, simulation replication latency (time between human movement and simulated arm response), and dataset completeness (percentage of frames with all target joints detected across all cameras).

The dataset generated by the system will be structured and documented for public or academic release, enabling other researchers and practitioners to use it directly for robotic arm training without requiring their own motion capture infrastructure. This positions MotionBridge as both a working system and a data contribution to the robotics research community.

A. APPENDICES

A.1. PROJECT BUDGET

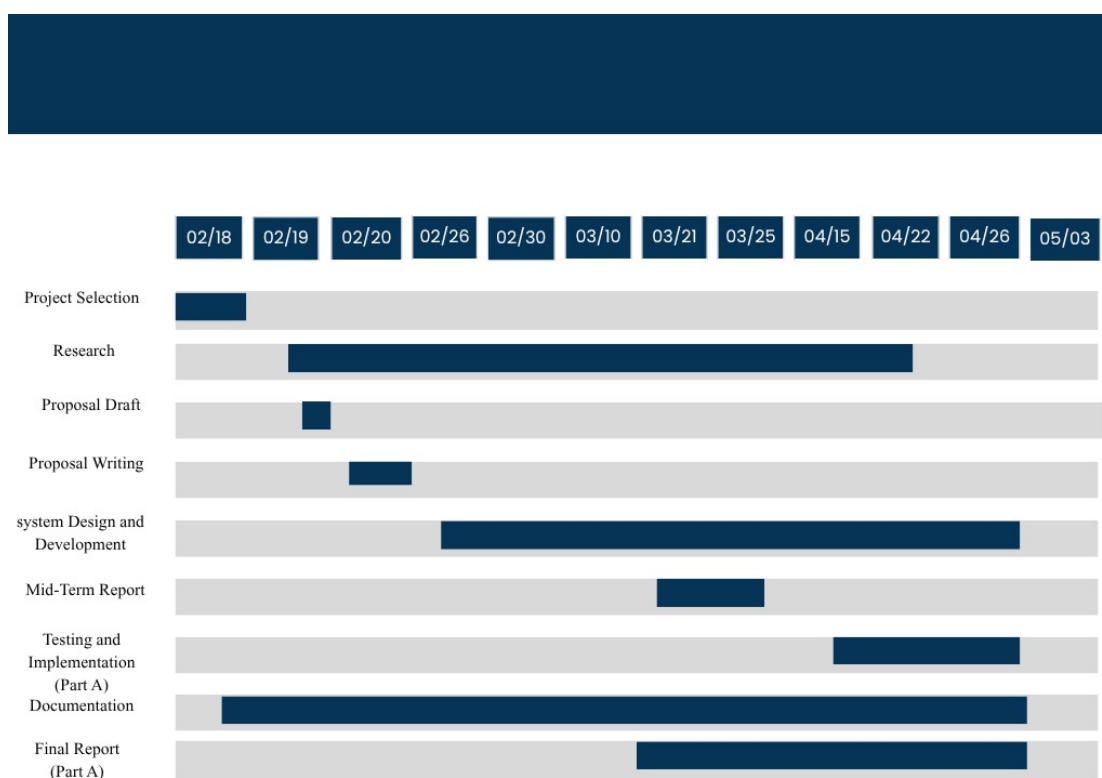
Table A-1. Estimated Project Budget for MotionBridge

S.N.	Qty	Unit Price (NPR)	Total Price (NPR)	
1	USB Webcams	3	4000	12000
2	Camera Tripods	3	1,500	4,500
3	Powered USB Hub	1	2,000	2,000
4	Calibration Target	1	1,000	1,000
5	Software & Tools	-	-	Free
6	Project Documentation	-	2,500	2,500
7	Contingency Fund	-	2,000	2,000
			Total Estimated Budget:	NPR 24,000

A.2. PROJECT TIMELINE

A.2.1. Gantt Chart

Table A-2. Gantt Chart for Project Timeline



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